

ZACHARY OBISPO

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Summary

Senior Aerospace Engineering undergraduate student with a strong interest in fluid dynamics and hands-on rocketry experience. Proficient in computational flow simulations and 3D CAD modeling via **ANSYS** and **SOLIDWORKS**. Seeking opportunities to apply CFD algorithms to research engineering challenges involving fluid flow and aerodynamic systems.

Education

University of California, San Diego

Sep. 2023 – June 2027 (expected)

Bachelor of Science in Aerospace Engineering

La Jolla, CA

GPA: 3.72

Relevant Coursework

- Advanced Fluid Mechanics
- Aerospace Materials/Structures
- Manual and CNC Machining
- Computational Fluid Dynamics
- MATLAB Programming
- Linear Controls

Experience

Triton Droids — Mechanical Team Member

August 2026 – Present

*Improved the torso of a humanoid robot via **Fusion 360** alongside a team of engineers*

La Jolla, CA

- Reduced weight and improved structural integrity by switching to an open cage design comprised of T-slotted frames
- Ensured compatibility with limbs and internal systems at vital connection points through collaboration with other teams
- Preserved design decisions and reasoning by documenting component geometry with precise measurements.

Shear Flow Instability Research — Undergraduate Researcher

April 2026 – Present

*Studied laminar mixing layers via **ANSYS Fluent** with oversight from MAE Prof. Keiko Nomura*

La Jolla, CA

- Validated results by comparing vorticity, velocity, and growth rate plots to published academic articles
- Demonstrated that finite-amplitude disturbances are required to trigger the growth of unsteady mixing-layer instabilities
- Minimized computational cost through a mesh study that reduced runtime without compromising accuracy

Rocket Propulsion Laboratory — Project Engineer

October 2024 – May 2025

*Designed a rocket powered by a G-class motor using **SOLIDWORKS** and **OpenRocket***

La Jolla, CA

- Optimized component geometry via flow/pressure simulations by identifying failure points and minimizing drag
- Programmed **Arduino**-powered avionics to collect real-time altitude, pressure, and temperature
- Successfully launched, achieving an altitude of 2500 ft, and safely recovered with minimal damage

County of San Diego, Dept. of Child Support Services — Student Worker

August 2024 – Present

Located absent parents to develop best practices for training while also impacting caseloads

Kearny Mesa, CA

- Followed **confidentiality** standards to maintain individuals' personally identifiable information
- Collaborated with fellow team members to optimize the time consumed for each locate task
- Assisted colleagues with their tasks, such as organizing training spaces, digital media production, and video editing

Projects

CFD Algorithm Development

May 2026 – May 2026

*Created CFD code via **MATLAB** to simulate 2D supersonic viscous flow over a stationary flat plate*

La Jolla, CA

- Implemented MacCormack's Method to iteratively solve the steady-state solution of air as an ideal gas flowing at Mach 4
- Validated the formation of an oblique shockwave by comparing visualized contours to verified simulations
- Communicated expectations alongside two fellow students and cross-verified code sections for accuracy and optimization

Solid Mechanics Problem Generator — MATLAB App Developer

January 2026 – June 2026

*Developed a **MATLAB** app to generate solid mechanics cases under MAE Prof. Marko Lubarda*

La Jolla, CA

- Engineered algorithms to apply multiple load cases for statically determinate systems with automated diagrams
- Located errors, improved reliability and accuracy through debugging, and optimized code structure using AI tools

Technical Skills

Software: ANSYS, MATLAB, SolidWorks, Fusion360, OpenRocket

Manufacturing: Manual and CNC Machining